Provider History & Sync Schema Plan

CurrencyRates - branch feat/core - APIs probed live on 2026-11-11

This document has two parts. Part 1 records what each integrated rate provider offers for historical data: how far back it goes, which intervals it supports, the endpoint to use and its request limits. Part 2 is the database plan for a provider catalog (used for filtering and options) and a sync configuration table that can only reference combinations the catalog allows.

Where a limit is not documented by the provider, the value comes from live API calls made on 2026-11-11 and may change.

Part 1 - Provider API integration requirements

1.1 Summary

Provider	Type	History depth	Intervals	Fit for full backfill
Frankfurter v2	FX	1999-01-04 (ECB); NBG source from 2009-01-05	daily, week, month	Yes - one call per range
NBG	FX	~1996 (USD only early; full table by 2005)	daily	Yes - one call per day
ExchangeRate-API	FX	None on free endpoint	-	No - paid key required
Binance	Crypto	From pair listing (BTCUSDT 2017-08)	1s ... 1M (16 options)	Yes - paged, 1,000 per call
Coinbase v2 prices	Crypto	From ~2023-10 only	daily	Partial
Coinbase Exchange	Crypto	From listing (BTC-USD before 2016)	1m, 5m, 15m, 1h, 6h, 1d	Yes - paged, 300 per call
Kraken	Crypto	OHLC: last 720 candles only	1m ... 15d (9 options)	No - OHLC too shallow
CoinGecko	Crypto	Past 365 days (free/Demo)	daily (auto granularity)	No - one year only

1.2 FX providers

Provider	History endpoint	Limits & behaviour
Frankfurter v2 `fx/frankfurter.py`	`GET /v2/rates?base&quotes&from&to[&group][&providers]`	Returns the whole range in one call (27 years = 6,875 rows; no cap found). `group=week|month` supported; `group=year` returns 422. Monthly value = average of business days (verified). Blended source fills weekends; `providers=ECB` gives business days only. If `from` is a non-business day, the previous published day is returned. Our `get_history` already uses this endpoint. Source start dates: ECB 1999-01-04, NBU 1999-01-04, NBG 2009-01-05, BOC 2017-01-03.
NBG `fx/nbg.py`	`GET .../currencies/en/json/?date=YYYY-MM-DD`	One day per call; no range parameter. A weekend or holiday date returns the last published table (2026-11-08 Sun returned 2026-11-07), so consecutive days produce duplicates - store by `validFromDate`. 1996-01-10 returned a 1996-01-06 table with USD only. Rates are GEL per `quantity` units. Endpoint is undocumented.
ExchangeRate-API `fx/er.py`	Free: none (`open.er-api.com/v6/history` returns 404). Paid: `v6.exchangerate-api.com/v6/{KEY}/history/USD/YYYY/MM/DD`	Latest only on the free endpoint, refreshed daily. Historical data requires a Pro plan or higher. Keep as a live source only.

1.3 Crypto providers

Provider	History endpoint	Limits & behaviour
Binance `crypto/binance.py`	`GET /api/v3/klines?symbol&interval&startTime&endTime&limit`	Intervals: `1s 1m 3m 5m 15m 30m 1h 2h 4h 6h 8h 12h 1d 3d 1w 1M`; anything else returns `-1120 Invalid interval`. Max 1,000 candles per call - a larger `limit` is silently capped, so page with `startTime`. BTCUSDT monthly data starts 2017-08. Quotes are in stablecoins (USDT), not USD. Rate limit: 418/429.
Coinbase v2 `crypto/coinbase.py`	`GET /v2/prices/{BASE}-{QUOTE}/spot?date=YYYY-MM-DD`	Daily spot only, one day per call. Data starts between 2023-09-01 and 2023-10-01; earlier dates return `rate not found` (undocumented). Fiat pairs like BTC-GEL were not found historically. Only `kind=spot` supports `date`.

Provider	History endpoint	Limits & behaviour
Coinbase Exchange (not integrated yet)	`GET api.exchange.coinbase.com/products/{pair}/candles?granularity&start&end`	Public, no key. Granularity (seconds): 60, 300, 900, 3600, 21600, 86400; others return `Unsupported granularity`. Max 300 candles per request (else error). BTC-USD daily candles exist for 2016. The deep-history option for Coinbase.
Kraken `crypto/kraken.py`	`GET /0/public/OHLC?pair&interval&since`	Intervals (minutes): 1, 5, 15, 30, 60, 240, 1440, 10080, 21600; others return `EGeneral:Invalid arguments`. Always returns only the last 720 candles, whatever `since` is: daily reached 2024-11-20 only, 15-day reached 2013. Full history requires paging `/0/public/Trades?since=` (data from 2013) and aggregating ourselves - slow.
CoinGecko `crypto/coin_gecko.py`	`/coins/{id}/history?date=dd-mm-yyyy` `/coins/{id}/market_chart/range`	Free and Demo plans: past 365 days only (error 10012). `history` returns one 00:00 UTC snapshot per call. `market_chart` granularity is automatic: ~5 min for 1 day, hourly for 2-90 days, daily beyond 90. Keyless ~5 calls/min. Supports fiat quotes such as GEL. Older data needs a paid plan.

1.4 Requirements for the historical sync

- **Intervals in the domain.** `IntervalKind` has DAILY and MONTHLY. Native support: Frankfurter (both), Binance (both), NBG / Coinbase / Kraken / CoinGecko (daily only - monthly must be derived from daily values).
- **Fetch by range, not by day.** `HistoricalRateSyncService.sync_range` makes one request per day: 27 years of Frankfurter would be ~10,000 calls instead of one. The port should gain a range method, e.g. `get_history(base, quote, start, end, interval)`, that each provider implements with its own paging.
- **Paging per provider.** Binance 1,000 / Coinbase Exchange 300 / Kraken 720 candles per call; NBG and Coinbase v2 one day per call; Frankfurter unlimited.
- **History start.** Some providers have a fixed earliest date (Frankfurter, NBG, Binance listing, Coinbase v2); others a rolling window (CoinGecko 365 days, Kraken ~720 daily candles). Sync configuration must be validated against both kinds.
- **Duplicates on non-business days.** NBG and Frankfurter return the previous published value for weekends and holidays; idempotent storage keyed by the provider's own date is required.
- **Not usable for backfill without changes:** ExchangeRate-API (paid key), CoinGecko (1 year), Kraken OHLC (720 candles), Coinbase v2 (from 2023-10).

Part 2 - Infrastructure DB plan

Four catalog tables describe what each provider can do; one configuration table describes what to sync. Composite foreign keys make the database reject a configuration the provider cannot serve. All tables build on `core.database.orm.Base` (UUID id), `AuditedBase` and `VersionMixin`.

2.1 Provider catalog

currencies - global list, so the UI can ask which providers support a currency

Column	Type	Notes
code	String(10), unique	USD, GEL, BTC, USDT
name	String(64)	
kind	enum fiat | crypto | stablecoin	for filtering

providers - one row per integration

Column	Type	Notes
key	String(40), unique	equals provider `name` and `RateRow.source` (nbg, frankfurter, binance ...)
display_name	String(64)	
kind	enum fx | crypto	
requires_api_key	bool	true only for ExchangeRate-API history
is_active	bool	turn a provider off without deleting its configuration

provider_intervals - capability per interval (limits differ per interval)

Column	Type	Notes
provider_id	FK -> providers	
interval	IntervalKind (DAILY / MONTHLY)	
is_native	bool	false = calculated from daily values (NBG monthly, Coinbase monthly)
history_start	Date, nullable	fixed earliest date (Frankfurter 1999-01-04, Coinbase v2 2023-10)
history_window_days	Integer, nullable	rolling limit instead of a fixed date (CoinGecko 365, Kraken ~720)
max_points_per_request	Integer, nullable	page size (Binance 1,000; Coinbase Exchange 300; null = unlimited)
	unique (provider_id, interval)	target of the composite FK from sync_configurations

provider_currencies - what each provider can quote

Column	Type	Notes
provider_id	FK -> providers	
currency_code	FK -> currencies.code	
provider_code	String(40), nullable	provider's own code: Kraken XBT, CoinGecko bitcoin (Python dicts today)
available_from	Date, nullable	e.g. NBG has only USD before 2005
	unique (provider_id, currency_code)	target of the composite FKs from sync_configurations

Per-currency rows are preferred over a pairs table: FX providers calculate cross rates for any pair, so a pairs table would explode. Binance / Kraken pair validity (e.g. whether BTCGEL exists) is checked when a sync configuration is created. Add a `provider_pairs` table later only if needed.

2.2 Sync configuration

sync_configurations (AuditedBase + VersionMixin)

Column	Type	Notes
provider_id	FK -> providers	
base, quote	String(10)	
interval	IntervalKind	
date_from	Date	Date, not datetime - both intervals are day-based
date_to	Date, nullable	null = open-ended: keep syncing new days
synced_until	Date, nullable	progress marker so an interrupted backfill resumes
is_active	bool	
options	JSON, nullable	provider-specific settings (Frankfurter providers=ECB, Coinbase kind)
last_run_at, last_status, last_error		for operators and the API

Constraints

- Composite FK `(provider_id, interval)` -> `provider_intervals`: no monthly config on a provider without monthly.
- Composite FKs `(provider_id, base)` and `(provider_id, quote)` -> `provider_currencies`: no unlisted currencies.
- Unique `(provider_id, base, quote, interval)`.
- Check `base <> quote` and `date_to IS NULL OR date_from <= date_to`.
- Partial index on `is_active` for the worker's polling query.
- Validation of `date_from` against `history_start` / `history_window_days` lives in the application service (the rolling window depends on today's date).

2.3 Repositories and domain

- `ProviderRepository`: `list(kind=, interval=, currency=)` for filters and options; `get_by_key`.
- `SyncConfigurationRepository`: `list_due()` (active, not finished), `advance(id, synced_until)` with optimistic locking, `record_failure`.
- Domain: replace `SyncConfiguration` / `TickerSyncConfig` with `ProviderCapability` and a `SyncConfig` value object.

2.4 Migration 0002_provider_catalog_and_sync_config

Create tables in order: currencies -> providers -> provider_intervals -> provider_currencies -> sync_configurations. Seed providers and intervals in the migration (fixed facts); load currencies with a refresh job or script (Frankfurter /v2/currencies, the NBG table, Binance exchangeInfo) because those lists change.

Provider	Daily	Monthly	Start / window	Page size
frankfurter	native	native (group=month)	1999-01-04	unlimited
nbg	native	derived	1996-01-06	1 day
binance	native	native (1M)	listing date	1,000
coinbase	native	derived	2023-10-01	1 day (v2)
kraken	native	derived	720-day window	720
coingecko	native	derived	365-day window	-
exchangerate-api	-	-	latest only	-

2.5 Fix first - existing model bugs

- `infrastructure/db/models/sync_configuration.py` imports `infrastructure.db.models.base` (does not exist) and uses dataclass `field()` instead of `mapped_column`, so it is not mapped.
- `core/database/orm.py` calls `uuid7` without importing it.
- `Base` adds an `id` primary key while `RateRow` declares a composite primary key - two primary-key definitions on `rates`. Drop `id` from `Base` or move it into a mixin.
- `domain/models.py`: `SyncConfiguration` references itself.